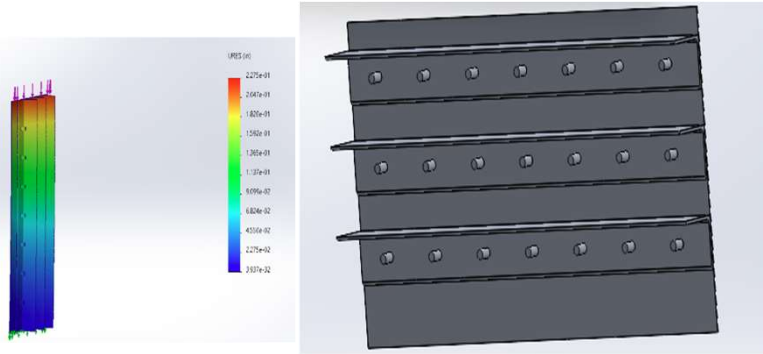


1. Stiffened Aluminum Panel Design

A comprehensive study is performed using analytical and numerical simulations to determine the structural performance of a stiffener-reinforced aluminum panel when a 3000lb load is applied.

4. Verification/Assembly

Upon finding the necessary rivet spacing and number of stiffened panels, I created an assembly in SolidWorks to run a stress simulation. The model and its simulation results are below:



5. Testing

Upon receiving these results in SolidWorks, I was confident with my design and decided it was ready to be tested under real stress. I created the stiffened panel in the senior design laboratory at NC state with help from a lab monitor. Upon finishing construction, I found a YouTube video online to install a strain gauge correctly and hook it up to a compression machine. Pictured to the right:



2. Nomenclature

a = Stiffener side length
 g = Gerard method number
 w = Effective sheet width
 C = End fixity coefficient
 p = Rivet spacing
 n = Number of stiffeners
 σ_{cs} = Crippling stress
 y = Compression yield stress
 σ_{ir} = Inter-rivet buckling stress
 E = Young's modulus of elasticity
 C_e = Needham Method coefficient
 l = Applied load
 ϵ = Axial strain
 A = Cross section area
 t = Structure sheet thickness

3. Theory

To accurately describe the stiffened panels effectiveness, I used the Needham and Gerard Methods prescribed below to determine the number of panels necessary and the required spacing of rivets to prevent serious deformation for the aluminum sheet.

Needham Method: $\frac{\sigma_{cs}}{\sqrt{\sigma_{cy}E}} = \frac{C_e}{t^{0.75}}$ Gerard Method: $\frac{\sigma_{cs}}{\sigma_{cy}} = 0.56 \left[\frac{gt^2}{A_{stiff}} \sqrt{\frac{E}{\sigma_{cy}}} \right]^{0.85}$

After finding the stiffened panel cross section, the crippling strength of aluminum and approximating the Needham Method coefficient through trial and error, I found the rivet spacing to be 1.5 inches based on the equation below:

$$\sigma_{ir} = \frac{\pi^2 Et}{\left(\frac{p}{0.58t}\right)^2}$$

6. Results

After running the compression machine, I was able to extract actual strain data from the strain gauge which I plotted with the strain data from my simulation in SolidWorks. as seen to the right:

